

LECTURE NOTES MATH255

SVEN SANDFELDT

1. LECTURE 14

In this class we go through some solutions to old problems in mathematics by using ideas of field extensions.

1.1. Solution to midterm problem. We begin by proving that if $f(x) \in \mathbb{Q}[x]$ has degree n and $a \in \mathbb{C}$ then $f(x)$ satisfies the formula:

$$(1.1) \quad f(x) = \sum_{j=0}^n \frac{f^{(j)}(a)}{j!} (x-a)^j.$$

We begin by noting that for $j \geq 1$ we have:

$$\begin{aligned} x^j &= (x-a+a)^j = \sum_{i=0}^j \binom{j}{i} a^{j-i} (x-a)^i = \\ &= \sum_{i=0}^j \frac{j(j-1)(j-2)\dots(j-i+1) \cdot a^{j-i}}{i!} (x-a)^i = \sum_{i=0}^j \frac{(x^j)^{(i)}(a)}{i!} (x-a)^i, \end{aligned}$$

where we have used that $(x^j)^{(i)} = j(j-1)(j-2)\dots(j-i+1)x^{j-i}$. That is, the formula holds for x^j . Consider now an arbitrary polynomial of degree n :

$$\begin{aligned} f(x) &= \sum_{j=0}^n a_j x^j = \sum_{j=0}^n a_j \sum_{i=0}^j \frac{(x^j)^{(i)}(a)}{i!} (x-a)^i = \\ &= \sum_{i=0}^n \sum_{j=0}^n \frac{a_j (x^j)^{(i)}(a)}{i!} (x-a)^i = \\ &= \sum_{i=0}^n \frac{\left(\sum_{j=0}^n a_j x^j\right)^{(i)}(a)}{i!} (x-a)^i = \\ &= \sum_{i=0}^n \frac{f^{(i)}(a)}{i!} (x-a)^i \end{aligned}$$

which is what we wanted to prove.

Next we show that if $f(x) \in \mathbb{Q}[x]$ has a zero of degree $r \geq 2$ at $\alpha \in \mathbb{C}$ then $f'(x)$ also has a zero at α . Note that this is an immediate consequence of the product rule, but

we will use the previous part of the problem. Let α be a zero of $f(x)$ with multiplicity r then we can write using Taylors formula:

$$(1.2) \quad \sum_{j=0}^n \frac{f^{(j)}(\alpha)}{j!} (x - \alpha)^j = (x - \alpha)^r g(x)$$

evaluating at $x = \alpha$ the right hand side is zero, so we obtain that the term $f(\alpha) = 0$ (which we already knew), and we can write:

$$(1.3) \quad \sum_{j=1}^n \frac{f^{(j)}(\alpha)}{j!} (x - \alpha)^j = (x - \alpha) \left(\sum_{j=1}^n \frac{f^{(j)}(\alpha)}{j!} (x - \alpha)^{j-1} \right) = (x - \alpha)^r g(x)$$

and if we divide by $x - \alpha$ on both sides (recalling that $\mathbb{Q}[x]$ is an integral domain) we obtain:

$$(1.4) \quad \sum_{j=1}^n \frac{f^{(j)}(\alpha)}{j!} (x - \alpha)^{j-1} = (x - \alpha)^{r-1} g(x)$$

So, if $r \geq 2$ then we can evaluate at $x = \alpha$ and the left hand side becomes $f'(\alpha)$ while the right hand side is 0, so $f'(\alpha) = 0$.

Finally, let $f(x) \in \mathbb{Q}[x]$ be irreducible, and $\alpha \in \mathbb{C}$ a zero for $f(x)$ with multiplicity $r \geq 2$. It follows that $f'(x) \in \mathbb{Q}[x]$ has a zero at α as well, so since $f(x)$ is irreducible we conclude that $f(x)$ divides $f'(x)$. On the other hand, $f'(x)$ has degree less than $f(x)$ so this would imply $f'(x) = 0$ or that $f(x)$ is constant. But this is a contradiction, since irreducible polynomials are not constant.

1.2. Algebraic closure of the rationals. We now complete the proof from last class that

$$(1.5) \quad \overline{\mathbb{Q}} = \{\alpha \in \mathbb{C} \mid f(\alpha) = 0, \text{ for some non-constant } f(x) \in \mathbb{Q}[x]\}.$$

is the algebraic closure of $\mathbb{Q}$. Note that $\mathbb{Q} \subset \overline{\mathbb{Q}}$ since $q \in \mathbb{Q}$ is a zero of $x - q \in \mathbb{Q}[x]$, and as we saw last time $\overline{\mathbb{Q}}$ is a field. So $\overline{\mathbb{Q}}$ is an extension of $\mathbb{Q}$, and this extension is algebraic since we only add elements to $\overline{\mathbb{Q}}$ that are zeros of polynomials with coefficients in $\mathbb{Q}$. It remains to prove that $\overline{\mathbb{Q}}$ is algebraically closed.

Theorem 1.1. *The algebraic numbers $\overline{\mathbb{Q}}$ is an algebraically closed field.*

Proof. We need to show that if $f(x) \in \overline{\mathbb{Q}}[x]$ is non-constant then $f(x)$ has a zero in $\overline{\mathbb{Q}}$. Let $f(x) = c_n x^n + \dots + c_0 \in \overline{\mathbb{Q}}[x]$ where $c_n, \dots, c_0 \in \overline{\mathbb{Q}}$. Since each $c_0, \dots, c_n \in \overline{\mathbb{Q}}$ is algebraic over $\mathbb{Q}$ (since $\overline{\mathbb{Q}}$ is an algebraic extension) we can define the finite extension:

$$(1.6) \quad E = \mathbb{Q}(c_0, \dots, c_n)$$

of $\mathbb{Q}$. We note that $f(x) \in E[x]$. If $\alpha \in \mathbb{C}$ is a zero of $f(x)$ (note that $\mathbb{C}$ is algebraically closed so such α exists) then α is algebraic over E , so $E(\alpha)$ is a finite extension of E . From the tower property it follows that $E(\alpha)$ is a finite extension of $\mathbb{Q}$ (i.e., $[E(\alpha) : \mathbb{Q}] = [E(\alpha) : E] \cdot [E : \mathbb{Q}] < \infty$). Since any finite extension of $\mathbb{Q}$ is algebraic we conclude that $E(\alpha)$ is an algebraic extension of $\mathbb{Q}$, and since $\alpha \in E(\alpha)$ we conclude

that α is algebraic over $\mathbb{Q}$ or $\alpha \in \overline{\mathbb{Q}}$. That is, if $f(x) \in \overline{\mathbb{Q}}[x]$ is any non-constant polynomial then $f(x)$ has a zero α that is in $\overline{\mathbb{Q}}$, so $\overline{\mathbb{Q}}$ is algebraically closed. ■

1.3. Geometrically constructible numbers. In this section we will consider an application of field extensions to straight edge and compass constructions in Euclidean geometry. Consider the plane $\mathbb{R}^2$. The constructible numbers $F \subset \mathbb{R}$ is every real number α whose norm $|\alpha|$ can be constructed as the length of line segment obtained using our straight edge and compass starting with the two points in the plane $(0, 0)$ and $(1, 0)$. For example, if we place the center compass at $(1, 0)$ and the edge at $(0, 0)$ and draw the circle, then the circle intersects the x -axis in $(0, 2)$ and then we can draw the line segment between $(0, 0)$ and $(0, 2)$ using the straight edge to produce a line of length 2, so $2 \in F$. We refer to the book for the following theorem:

Theorem 1.2. *The constructible numbers F form a subfield of $\mathbb{R}$ that contains $\mathbb{Q}$.*

Since we can construct all rational numbers we can give a description of the points in the plane that we can produce as follows:

- (i) We can take two lines ℓ_1 and ℓ_2 that are not parallel, passing through rational points and take their intersection,
- (ii) we can draw a circle around centered at a rational point and with rational radius and intersect it with a line passing through two rational points,
- (iii) we can take the intersection points of two circles with rational centers and with rational radii.

We can then keep using (i), (ii), and (iii) but at each step we start with the points obtained in the previous step instead of the rationals. Note that the intersection between two lines passing through rational points is a rational point in $\mathbb{R}^2$, so this does not produce any new points in the plane. However, if we take the intersection point of a line $ax + by + c = 0$ and a circle $x^2 + y^2 + dx + ey + f = 0$ then we can solve for y in the first equation: $y = -c - ax$ and put this in the second equation:

$$(1.7) \quad x^2 + (c + ax)^2 + dx - e(c + ax) + f = 0$$

which is a quadratic equation for x . By the quadratic formula we deduce that x can be obtained by first adding and multiplying some rational numbers, then taking a square root and finally adding a rational number. This is the motivation for the following theorem (for more details see the book!).

Theorem 1.3. *Let $\gamma \in F$ be a constructible number. Then there are numbers $\alpha_1, \alpha_2, \dots, \alpha_n = \gamma$ such that $\mathbb{Q}(\alpha_1, \dots, \alpha_i)(\alpha_{i+1})$, $i < n$, is an extension of $\mathbb{Q}(\alpha_1, \dots, \alpha_i)$ of degree 2 (obtained by taking a square root of some element in $\mathbb{Q}(\alpha_1, \dots, \alpha_i)$).*

As a consequence we have the following:

Theorem 1.4. *Let $\gamma \in F$ be a constructible number. Then $[\mathbb{Q}(\gamma) : \mathbb{Q}] = 2^r$ for some $r \geq 1$.*

Proof. Let $\alpha_1, \dots, \alpha_n = \gamma$ be as in the previous theorem. By the tower property of degrees (and induction) we have

$$\begin{aligned} [\mathbb{Q}(\alpha_1, \dots, \alpha_n) : \mathbb{Q}] &= [\mathbb{Q}(\alpha_1) : \mathbb{Q}] \cdot [\mathbb{Q}(\alpha_1, \alpha_2) : \mathbb{Q}(\alpha_1)] \cdot \dots \cdot \\ &\quad [\mathbb{Q}(\alpha_1, \dots, \alpha_n) : \mathbb{Q}(\alpha_1, \dots, \alpha_{n-1})] = 2^n. \end{aligned}$$

Again using the tower property we have $2^n = [\mathbb{Q}(\alpha_1, \dots, \alpha_n) : \mathbb{Q}] = [\mathbb{Q}(\gamma) : \mathbb{Q}] \cdot [\mathbb{Q}(\alpha_1, \dots, \alpha_n) : \mathbb{Q}(\gamma)]$ so since the only prime factor on the left is 2 we conclude that $[\mathbb{Q}(\gamma) : \mathbb{Q}] = 2^r$ for some $r \geq 1$. ■

We can now obtain some immediate consequences of this theorem.

Theorem 1.5. *There exists a cube such that one can not construct, using straight edge and compass, a cube of twice the volume of the original cube. More precisely, given the length of the side of a cube it is not always possible with a straight edge and compass to produce the length of a cube with twice the volume.*

Proof. Consider the cube with side of unit length 1. A cube with twice its volume would have side length $2^{1/3}$. However, $2^{1/3}$ is not constructible since $[\mathbb{Q}(2^{1/3}) : \mathbb{Q}] = 3$ which is not a power of 2 (note that the irreducible polynomial of $2^{1/3}$ is $x^3 - 2$). ■

Theorem 1.6. *One can not square the circle in general. That is, given a circle it is not always possible to construct with straight edge and compass a square with the same area as the disk enclosed by the circle.*

Proof. Consider the circle with radius one, the enclosed disc has area π . So we would need to construct a square with side length $\sqrt{\pi}$. But note that $\sqrt{\pi}$ is not algebraic since π is not algebraic. Since any constructible number is algebraic (since the degree of any constructible number over $\mathbb{Q}$ is finite), it follows that it is impossible to produce a square with the same area as the disk enclosed by a unit radius circle using only a straight edge and a compass. ■

Email address: sandfeldt@uchicago.edu